MODULE 39 · SECTION 5 - MAINTAINING AND AUTOMATING DATA WORKLOADS

# Replication, Failover and Regional Outages

RPO and RTO, Cloud SQL HA and cross-region replicas, Spanner configurations, Bigtable replication, dual-region buckets, BigQuery managed disaster recovery, and Memorystore.

SECTION 5 of 5 EXAM WEIGHT 18% MODULE 39 of 45

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Two numbers decide every answer. How much data can you lose, and how long can you be down?

# RPO and RTO

RPO

## Recovery point objective

- How much data you can lose
- Measured back from the failure
- Set by how replication works
- Synchronous can reach zero

RTO

## Recovery time objective

- How long you can be down
- Measured forward from the failure
- Set by how failover works
- Automatic beats a manual runbook

REMEMBER

High availability survives a zone, usually automatically. Disaster recovery survives a region, often by hand.

# What already survives losing a zone

| SURVIVES A ZONE BY DEFAULT | NEEDS YOU TO ADD IT |
| --- | --- |
| **BigQuery**: two zones, always | **Cloud SQL**: turn on HA |
| **Cloud Storage** regional bucket | **Bigtable**: add a second cluster |
| **Spanner** regional: three zones | **Memorystore**: Standard tier |
| **Dataflow** with regional placement | **Spark** clusters: one zone each |

THE TRAP

Nothing on this slide survives losing a whole region. That always takes a deliberate choice.

# High availability: the zone layer

- A **standby** in a second zone. A write commits only once **both zones** have it.
- Failover is **automatic**, typically about **60 seconds**, on the same IP address.
- Costs **twice** a standalone instance: CPU, memory and storage.
- Same region only. HA does **nothing** for a regional outage.

> THE NUMBERS
>
> HA gives an RPO of zero and an RTO of about a minute, for a zone.

# Cross-region replicas: the region layer

- A read replica in **another region**, kept up to date **asynchronously**.
- In an outage, the last transactions may **not have arrived**. RPO is not zero.
- **Promote** it by hand. It becomes a standalone, writable primary.
- **Enterprise Plus** advanced DR: a designated DR replica, failover and switchover back.

> THE PATTERN
>
> Before promoting, check replication lag. Afterwards, turn on HA and recreate the other replicas.

# Larkspur's tariff database: the region runbook

```
# London is down. The DR replica is in Belgium, created in module 21.
gcloud sql instances describe lk-tariffs-dr --format='value(state)'

# 1. Promote it: now a standalone, writable primary
gcloud sql instances promote-replica lk-tariffs-dr

# 2. Give the new primary its own zone protection
gcloud sql instances patch lk-tariffs-dr --availability-type=REGIONAL

# 3. Point the tariff service at lk-tariffs-dr, then rebuild replicas
```

5 Manual and deliberate. There is no automatic regional failover here.

8 Regional means HA, in europe-west1 now.

FOR LARKSPUR

Rehearse this runbook. An untested DR plan has an unknown RTO.

# Three ways to place a Spanner instance

| CONFIGURATION | WHERE | AVAILABILITY |
|---|---|---|
| Regional | Three zones, one region | 99.99% |
| Dual-region | Two regions, one country | 99.999%, zero RPO |
| Multi-region | Two read-write regions, plus a witness | 99.999% |

CURRENCY CHECK

Dual-region and multi-region need the Enterprise Plus edition. Writes cost a little more latency.

# Choosing the configuration

- **eur5**: London leads, Belgium second, Netherlands as witness.
- **eur6**: Netherlands leads, Frankfurt second, Zurich as witness.
- **dual-region-germany1**: Berlin and Frankfurt. Both copies stay in Germany.
- No dual-region config pairs London with another UK region.

> THE TRAP
>
> Residency first, then availability. A multi-region config can cross the very border you must keep.

# Bigtable replication

- Add clusters in other zones or regions, up to **8 regions** per instance.
- Replication is **eventually consistent**, typically seconds to minutes behind.
- **App profiles** route traffic: single-cluster or multi-cluster routing.
- Conflicting writes to one cell: the **last write wins**.

> THE TIEBREAKER
>
> Need read-your-writes or strong consistency? Single-cluster routing. Fail over to another cluster only when needed.

# Two app profiles on meter_live

LIVE-APP

## Multi-cluster routing

- The customer app's reads and writes
- Nearest available cluster
- Fails over automatically
- Eventual consistency accepted

ANALYTICS

## Single-cluster routing

- Data science scans
- Pinned to the second cluster
- Never slows the customer app
- Reads a copy seconds behind

THE PAYOFF

One instance, two clusters, two profiles: availability and workload isolation from the same design.

# Buckets that survive a region

- **Regional**: at least two zones. Survives a zone, not a region.
- **Dual-region**: two regions you name. **Multi-region**: Google picks, 100 miles apart.
- Default replication: 99.9% within 1 hour, all within **12 hours**.
- **Turbo replication**: every object within **15 minutes**. Dual-region only.

> THE NUMBERS
>
> Durability is eleven nines everywhere. The location type changes availability and RPO, not durability.

# BigQuery across regions

- Every dataset keeps copies in **two zones** of its location, automatically.
- A **multi-region** location like EU does **not** add cross-region redundancy.
- **Dataset replication**: a read-only replica elsewhere. Promotion needs the primary online.
- **Managed disaster recovery**: compute and storage fail over. Enterprise Plus only.

> THE TRAP
>
> The docs say dataset replication is not a plan for losing a whole region. Managed disaster recovery is.

# Managed disaster recovery for the warehouse

```
1  -- 1. An Enterprise Plus failover reservation, standby in Belgium
2  CREATE RESERVATION `lk-admin-prod.region-europe-west2.warehouse_dr`
3  OPTIONS (slot_capacity = 200, edition = ENTERPRISE_PLUS,
4           secondary_location = 'europe-west1');
5
6  -- 2. Attach each dataset to it
7  ALTER SCHEMA curated SET OPTIONS (
8    failover_reservation = 'lk-admin-prod.warehouse_dr');
9
10 -- 3. In an outage, run in the secondary region
11 ALTER RESERVATION `lk-admin-prod.region-europe-west1.warehouse_dr`
12 SET OPTIONS (is_primary = TRUE, failover_mode = 'HARD');
```

3-4 Enterprise Plus, with a secondary location.

8 Each dataset is attached.

12 Hard does not wait. Soft waits, but needs both regions up.

THE NUMBERS

Capacity is ready five minutes after failover starts. The RPO goal is fifteen minutes.

# Memorystore failover

- **Basic tier**: one node, no replica. A failure empties the cache.
- **Standard tier**: a primary and a replica, with **automatic failover**.
- About **30 seconds** on average. Clients reconnect to the same IP.
- Replication is **asynchronous**, so acknowledged writes may be lost.

> REMEMBER
>
> A cache can always be rebuilt from the database. Never make it the only copy of anything.

# Phrases that pick the answer

| THE QUESTION SAYS | REACH FOR |
|---|---|
| Cloud SQL, survive a **zone** | High availability, regional instance |
| Cloud SQL, survive a **region** | Cross-region replica, then promote |
| **Zero RPO** across regions, relational | Spanner dual-region or multi-region |
| Bigtable, isolate **batch scans** | Second cluster, single-cluster profile |
| Bucket, every object within **15 minutes** | Dual-region with turbo replication |
| BigQuery, query and write after a **region loss** | Managed DR, Enterprise Plus |

THE PATTERN

Synchronous means zero loss and more cost. Asynchronous means a small loss and a manual step.

> The tariff database runs on Cloud SQL for PostgreSQL in London, with high availability turned on.
>
> The board wants it back in another region within an hour if London is lost. A few minutes of lost writes is acceptable, at the lowest cost.

## What should the team add?

A Nothing, because HA already fails over to a standby automatically

B A cross-region read replica, promoted by hand in an outage

C A move of the whole database to a Spanner multi-region instance

D A nightly export of the database to a bucket in another region

◆ PAUSE AND DECIDE

## What should the team add?

A Nothing more

*The standby is in the same region.*

B Cross-region replica

Minutes of loss, recovery well inside an hour.

C Spanner multi-region

*Zero loss, but a migration and far more cost.*

D Nightly export

*Up to a day of lost writes.*

THE SIGNAL

Survive a region, minutes of loss allowed: a cross-region replica.

> The customer app reads and writes meter_live in a single-cluster Bigtable instance in London.
>
> Data science scans of the same table slow the app at peak, and the app must also survive a zone failure.

## What should the team do?

A Add nodes to the existing cluster, and run the scans overnight

B Copy meter_live into BigQuery every hour for the data scientists

C Add a second cluster, and give both workloads one multi-cluster profile

D Add a second cluster: scans pinned to it, the app on multi-cluster

◆ PAUSE AND DECIDE

## What should the team do?

A More nodes

*Still one zone, and scans still compete.*

B Hourly copy

*Helps the scans, but not the zone failure.*

C One shared profile

*Scans can land on the app's cluster.*

D Two profiles

Isolation for scans, failover for the app.

THE SIGNAL

Isolate workloads and survive a zone: two clusters, two app profiles.

> Larkspur is building a billing ledger for German customers. It needs strong consistency and no lost transactions, even if a region is lost.
>
> Under the data protection agreement, every copy of the data must stay in Germany.

## Which design fits these needs?

A. Cloud SQL with HA in europe-west3, plus a replica in europe-west1

B. Spanner multi-region eur6, which has Frankfurt as a read-write region

C. Spanner dual-region-germany1, across Berlin and Frankfurt

D. Bigtable with clusters in Frankfurt and Berlin, on multi-cluster routing

◆ PAUSE AND DECIDE

## Which design fits these needs?

A Cloud SQL plus replica

*The replica leaves Germany, and lags behind.*

B Spanner eur6

*Copies in the Netherlands and Zurich.*

C Spanner dual-region

Zero RPO, and both regions are German.

D Bigtable

*Eventually consistent, and no transactions.*

THE SIGNAL

Zero loss across regions, one country: Spanner dual-region.

> Larkspur's warehouse runs on BigQuery Enterprise edition in London. A new rule says that if London is lost, analysts must query, and pipelines must write, from Belgium within minutes.
>
> Losing up to about fifteen minutes of changes is acceptable.

## What should the team set up?

A Managed DR on an Enterprise Plus failover reservation

B A dataset replica in europe-west1, promoted during the outage

C Datasets moved to the EU multi-region, which spans many regions

D A scheduled nightly copy of every dataset into europe-west1

◆ PAUSE AND DECIDE

## What should the team set up?

A **Managed DR**
Compute and storage fail over in minutes.

B **Dataset replica**
*Promotion needs London online.*

C **EU multi-region**
*No cross-region redundancy added.*

D **Nightly copy**
*A day of loss, and no compute ready.*

THE SIGNAL

Query and write after a region loss: managed DR, Enterprise Plus.

# Module 39 in five lines

- **RPO** is data you can lose. **RTO** is time you can be down. Read both first.
- Cloud SQL: **HA** for a zone, a **cross-region replica** promoted by hand for a region.
- Spanner dual-region or multi-region: **zero RPO**. Check where the regions are.
- Bigtable: clusters plus **app profiles**. Buckets: dual-region, **turbo** for 15 minutes.
- BigQuery multi-region is not DR. **Managed DR** needs Enterprise Plus.

COMING UP

Next: module 40, every rename and retirement, with the exam-safe answer.